Dequantization on Geometry

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I'm interested in the re-interpretation of non-linear PDE from algebraic geometry and QFT perspective. It's typically Navier-Stokes, Apery, period integral of Landau Ginzburg, Monge-Ampere below, and I consider possible strategies are 1: I consider non-linear PDE differential geometrically, by making some connection form diffyq $\nabla f = 0$ on a real smooth manifold X, possibly symplectification (X', ω) if necessary, where I further put geometric quantization, and 2: Derived moduli space of solutions spaces of non-linear PDE, whose algebraic geometric realization, and 3: Transform non-linear PDE to CFT, corresponding to algebraic geometry.

- Navier Stokes

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}$$

  Or

$$\partial_\mu T^{\mu\nu} = 0$$

- Apery (Quantum Satake)

$$\mathcal{L}_{\mathrm{Ap}} = \theta^3 - z(2\theta+1)(17\theta^2+17\theta+5) + z^2(\theta+1)^3, \qquad \theta = z\frac{d}{dz}.$$

- Landau-Ginzburg (Period Integral, SCFT)

$$\int e^{-f(x)} \qquad f(x) = \text{polynomial}$$

- Monge-Ampere (SYZ)

$$\det\left(\frac{\partial^2 \phi}{\partial x_i \partial x_j}\right) = f(x)$$

- SUGRA (String Theory)

$$R_{\mu\nu} - \frac{1}{2}R\, g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Remark 0.1

I also noticed several different approaches, Kcryzka introduces derived algebraic geometry realization of jet spaces, Ladychenskaya formula for solving 2d NSE. These are quite interesting but quite different from my study.

keywords:
picard-fuch, quantum connection.

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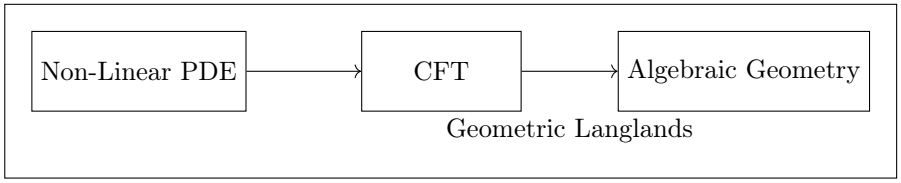
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1 Navier-Stokes



Proposition 1.1. NSE-CFT Conversion

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}$$

transformed into

$$\partial_\mu T^{\mu\nu} = 0$$

The question.

Question 1.1

- Connection form diffyq
- Symplectic extension
- Quantization

All real smooth manifold can be a Lagrangian submanifold of some Symplectic manifold. In particular cotangent bundle, $\iota : M \hookrightarrow T^*M$. How does this embedding maps connection form diffyq $\nabla f = 0$? In addition, connection ∇ on M needs to have natural extension $\tilde{\nabla}$ on T^*M, so its restriction is $\tilde{\nabla}|_M = \nabla$.

$$\begin{aligned} M &\hookrightarrow T^*M \\ \nabla &\mapsto \tilde{\nabla} \end{aligned} \tag{1}$$

I define quantization of non-linear PDE by quantization of connection form ∇ in symplectic geometry.

Definition 1.1

Definition 1.2. Quantization of NSE

2 Apery

The basic idea of quantum Satake correspondence is to configure quantum connection to solve a variety of quantum differential equations. Roughly speaking, generalization of Guass-Manin and Picard-Fuch.

Definition 2.1. Geometric Satake

$$\mathrm{Perv}(\mathrm{Gr}_G) \simeq \mathrm{Rep}(G^\vee).$$

Definition 2.2. Quantum Satake

Definition 2.3. 4D/2D CFT Correspondence

Definition 2.4

Definition 2.5

Definition 2.6

3 Quantization

One of my interests in non-linear PDE is to build framework of quantization, which is first of all embedding of connection form $\nabla f = 0$ to symplectic geometry, and secondly to do quantization, possibly Fedosov's method of geometric construction by deformation quantization, or geometric prequantization: these two are different, and I don't know which suits better. In any case, quantization of Navier-Stokes equation is consider to be difficult according to ChatGPT.

Definition 3.1. Geometric Prequantization

Prequantization is a process of making an operator $Q(f)$ from a smooth function f, $Q : C^\infty(X) \to End(\mathcal{H})$ on a symplectic manifold (X, ω). More precisely, $\mathcal{H} = \Gamma^\infty(L)$ where L is a prequantum line bundle with a connection ∇_{X_f}, and an operator given by $Q(f) := -i\hbar\nabla_{X_f} + f$, so that bracket satisfies $[Q(f), Q(g)] = i\hbar Q(\{f, g\})$.

Typically, the operator is Kostant–Souriau operator $Q(f) := -i\hbar(X_f + \frac{1}{i\hbar}\theta(X_f)) + f$.

Definition 3.2. Deformation Quantization

Deformation quantization transforms ordinary product to star product $fg \mapsto f \star g$ where the star product is defined by

$$f \star g = \Sigma_{n\geqq 0}\frac{1}{n!}(\frac{i\hbar}{2})^n \Pi^n(f, g)$$

where Π is Poisson bivector, and $\Pi^1(f, g) = \{f, g\}$ is ordinary bracket. Naturally, the Poisson bracket is

defined by

$$\frac{f \star g - g \star f}{i\hbar} = \{f, g\} + O(\hbar)$$

Fedosov's method of deformation quantization in symplectic geometry.

Definition 3.3. Fedosov's Deformation Quantization

TBA

Definition 3.4

4 Monge-Ampere

Monge Ampere is SYZ mirror symmetry.

Definition 4.1

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